Python and Google Colab — participant handout

BrainHack 2026 Workshop — Speech Recognition and Generation
Module 3 (10 min)

> A short introduction so that everyone can run code in the hands-on part. No prior programming experience is required.

1. What is Python?

Python is a programming language — one of the most popular in the world, especially in AI and data analysis.

- Easy to read and write (it resembles pseudocode).
- It has a huge ecosystem of libraries (e.g. for working with audio, text, AI).
- In the workshop we use it to call APIs and run models (e.g. Whisper).

Learning from scratch (free courses): https://learn.python.land/

2. What is Google Colab?

Google Colab is a Jupyter Notebook running in the Google cloud — a "notebook" with code cells that you run in your browser.

- Nothing to install — a Google account is enough.
- Free GPU/CPU for computations.
- You write and run code cell by cell (`Shift+Enter`).
- You can upload your own files (e.g. a recording) and download results.

Link: https://colab.research.google.com

3. Practice (during the workshop)

1. Download the repo from GitHub as a ZIP: on the repository page choose **Code → Download ZIP** and unpack the file. Then in Colab choose File → Upload notebook and point to `Local_ASR_models.ipynb` from the unpacked folder.
2. Run the first cell (`Shift+Enter`).
3. Upload your own recording (files.upload / mount Drive).

Links

- Colab: https://colab.research.google.com
- Colab FAQ (GPU): https://research.google.com/colaboratory/faq.html
- Python for beginners: https://www.python.org/about/gettingstarted/
- Python courses: https://learn.python.land/